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Hydrocarbon Prospectivity Assessment of the Ultra Deep Reservoirs of Kuwait

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Abstract

Triassic-Paleozoic reservoirs hydrocarbon potential is proven in the Arabian plate, however these reservoirs in The State of Kuwait exist at greater depths and this combined with the related geological complexities pose severe challenges for exploration and development. An attempt to identify and resolve these challenges is necessary to provide an extensive evaluation of these deeper reservoirs and produce viable prospects for exploration.

Limited number of wells targeting Triassic-Paleozoic reservoirs has led to challenges in understanding these deep reservoirs and their entrapments. Triassic-Paleozoic sequences in the seismic are affected with poor resolution, multiples and noise. The less penetration of energy has caused reduced signal to noise ratio affecting the proper imaging of primary events corresponding to Formation boundaries. The regional analogues indicate the occurrence of a complex geology in Triassic-Paleozoic sequences. A recent full azimuth 3D depth domain reprocessed seismic data over part of The State of Kuwait has brought out enhanced image of deeper sections. Seismic reservoir characterization utilizing simultaneous inversion, seismic attributes, spectral decomposition, and acoustic impedance in this newly reprocessed 3D seismic data volume deciphered reservoir properties in addition to better structural and stratigraphic mapping.

Assessment of the basement architecture and the tectono-stratigraphy, along with the development of the paleo depositional environment and analysis of the regional context of the Triassic-Paleozoic on the Arabian Plate will identify potential reservoir strata and entrapments in Kuwait. The existence of intra-basinal highs with the possibility of thick Paleozoic source rock deposition in the adjacent basinal lows can form the potential locales for exploration. Preliminary structural analysis indicates these are older structures controlled by basement tectonics. Furthermore, it is understood that late Jurassic time is the peak migration period for deeper hydrocarbon sources and any structuration that predates this period can provide the necessary entrapment in these structures.

The recent 3D seismic reprocessing has produced a better image in the Triassic-Paleozoic in this area of The State of Kuwait. The analysis of Gravity-Magnetic data to implement a deep structural model along with regional evaluation of tectono-stratigraphy and paleoenvironments supported the structural and stratigraphic interpretations in the Triassic-Paleozoic over this 3D seismic volume. The probability of

adequate source rock and petroleum system timing was established and implemented in these geological models. Seismic reservoir characterization has additionally identified potential leads within these structural and stratigraphic traps. Successful exploration of the prospects identified will provide significant impetus for future exploration and development.

With the advent of modern technologies and innovative exploration practices along with the regional understanding, challenges associated with the Triassic-Paleozoic in Kuwait are being addressed and resolved leading to a more refined subsurface model.

Introduction

The State of Kuwait is one of the most prolific hydrocarbon-rich countries in the Arabian Peninsula. Commercial hydrocarbons have been proven from almost entire geological succession, from Tertiary to Triassic. Tertiary, Cretaceous and Jurassic formations are the major contributors to commercial hydrocarbons; Triassic, although proven yet to be developed for commerciality. Hydrocarbon production from the deeper Triassic-Paleozoic successions has been proven in the Arabian plate and these formations are yet to be fully explored and developed in the State of Kuwait. Based on the geology and the regional analogues, Minjur, Jilh and Sudair formations, representing the Triassic section of Kuwait, are envisaged to have significant potential for non-associated gas and condensates. Commercial flow of hydrocarbons from Lower Jilh from more than one well has established the presence of active petroleum systems in these deeper formations. These reservoirs exist at greater depths, and this, associated with geological complexities, poses severe challenges for their exploration and development. The limitation on the number of wells drilled on different structures/prospects that have penetrated these deeper formations within the country has led to inadequate understanding of the sub-surface. Countrywide high- density full azimuth 3D seismic and airborne Gravity & Magnetic data are available for further understanding of the subsurface. The available data and the regional analogues over the region are considered to assess the petroleum system elements for these deep reservoirs. An attempt has been made to identify and address the challenges for an extensive evaluation of these deeper reservoirs to identify viable prospects for exploration.

Methodology

Triassic-Paleozoic formations in onland Kuwait are penetrated in a few of the wells, with a wide range of areal distribution. Although the wells drilled through Paleozoic are very less, the Triassic section is reasonably covered. Additionally, the recent re-processed high- density full azimuth 3D seismic in-depth domain over a selected area has produced a better image of the subsurface. Based on these a comprehensive subsurface assessment was carried out along with the petroleum system model and regional analogues. This study discusses the ultradeep Triassic formations in detail utilizing the data collected through wells (Fig. 1) and the newly available seismic data with a preliminary understanding of Paleozoic and older formations, in view of their hydrocarbon potential.

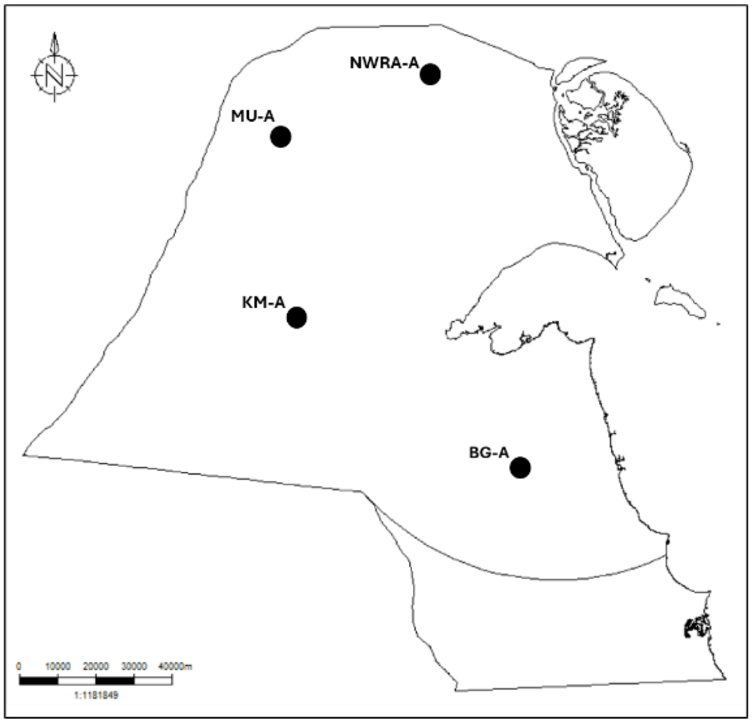

Figure 1—Location map showing the key wells in The State of Kuwait.

Triassic Stratigraphy and Plays

The Triassic strata form a part of the tectonostratigraphic mega sequence AP6 (Sharland et al. 2001). Davies & Simmons have provided a chronostratigraphic view of the Triassic formations in the regional chronostratigraphic context (Fig. 2).

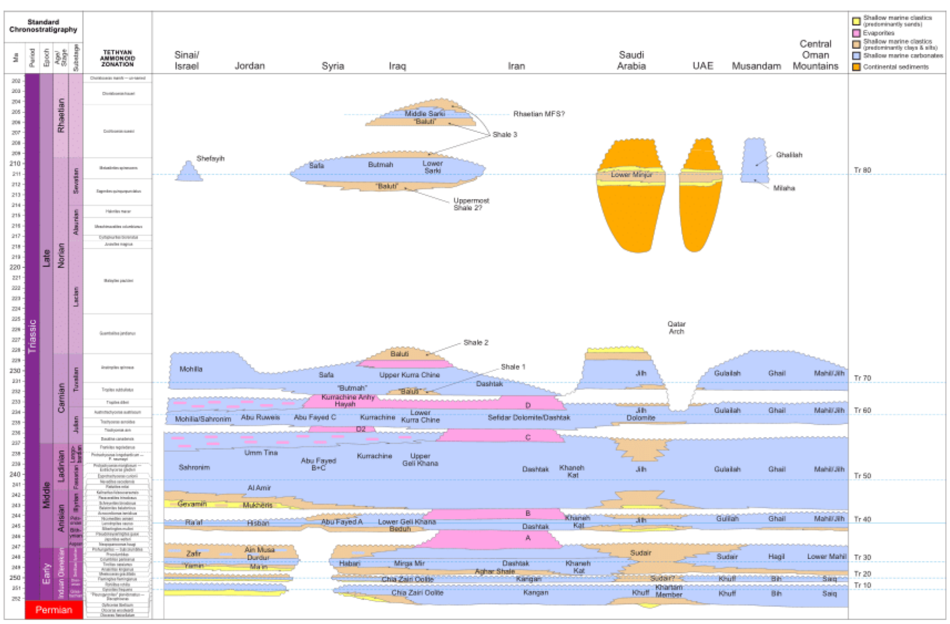

Figure 2—Regional Triassic chronostratigraphic chart (After Davies & Simmons).

The Triassic section comprises of four formations in Kuwait (Fig. 3) and a typical log response of these formations is provided.

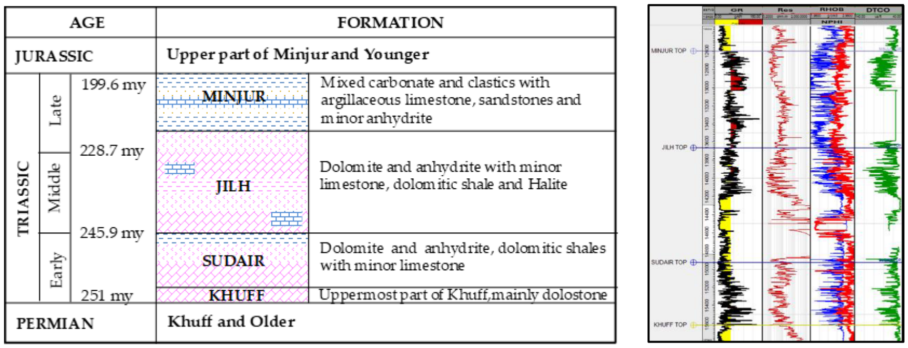

Figure 3—Generalized Triassic-Paleozoic stratigraphic column in Kuwait and the representative log motif from the drilled well.

These formations are:

- Minjur Formation of late Triassic to Early Jurassic age
- Jilh Formation of Middle Triassic age
- Sudair Formation of Early to Middle Triassic age
- Khuff Formation of Middle Permian to Early Triassic age

The Sudair Formation was deposited during Early Triassic (Induan-Olenekian) epoch (Haq and Qahtani, 2005) and attains a maximum thickness of 1,120 ft in Kuwait from the drilled well in KM. The formation conformably overlies the Khuff Formation and is in turn conformably overlain by the Jilh Formation. The formation was described by Khan (1989) in the well BG-A, divisible into Lower, Middle and Upper Sudair members.

Lower Sudair is encountered in the drilled wells KM-A, KM-B and MU-A (Fig. 4) and it dominantly is comprised of argillaceous dolostone grading to dolomitic shales and dolostone with anhydrites. The unit is dominantly dolomitic in the lower part and anhydritic in the upper part and its deposition appears restricted to the Dibdibba Trough. Middle Sudair is encountered in the drilled wells KM-B, MU-A, NWRA-A, NWRA-B and BG-A and dominantly consists of argillaceous dolostone with sub-ordinate anhydrites. This unit is observed in a relatively wider area in Dibdibba Trough as compared to Lower Sudair. Thickness of the lower and middle units increases in KM and MU area and the data from the wells drilled indicate lower porosity compared to the other areas. Additionally, the wells drilled on BG shows high variation in shale and anhydrite presence. The Upper Sudair is developed all over Kuwait and is made up dominantly of argillaceous dolostone and dolostone with minor anhydrite in Dibdibba Trough and dominantly evaporitic facies over Kuwait Arch with locally developed halites in BG-A. The wells drilled on MU and KM indicate low anhydrite percentage as compared to other wells with low porosity development.

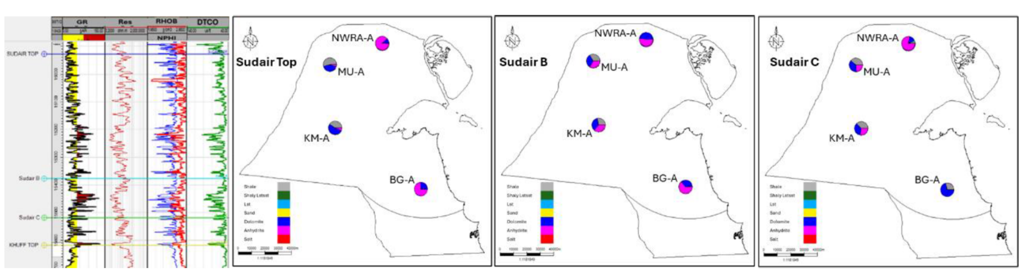

Figure 4—Log representation of Sudair section and the maps showing rock types derived based on petrophysical evaluation.

Based on the petrophyiscal assessment on the logs from the wells drilled, the deposition of the lower and the middle units in inferred to be at shallower environments compared to the upper unit (Fig. 4). Significant amounts of evaporites are observed in the lower and middle units, while a higher clay content is observed in the upper unit. In the northern part the evaporite presence is relatively higher. The biostratigraphic assessment also suggests, shallow conditions (transitional) prevailed over most of the studied area, with inner neritic conditions towards the south (Nicolas et. al 2024).

The Jilh Formation of Late - Middle Triassic (late Olenikian to early Norian) epoch deposited in shallow marine inner ramp and restricted settings is developed all over Kuwait with expanded section in western Kuwait. Haq and Qahtani (2005) have shown a major unconformity spanning from Anisian to Early Ladinian marking the Sudair-Jilh interface in Saudi Arabia. Khan (1989) described the formation in well BG-A, with a tripartite subdivision for the formation. The Lower and Upper Jilh are separated by Jilh Salt and Jilh Dolomite forming the Middle Jilh. The Lower Jilh is developed only in Dibdibba Trough and is drilled in KM-A, KM-B, MU-A and MU-B. The Lower Jilh comprises of a limestone rich unit in the basal part, a dominantly anhydrite and dolomite rich middle section and a dolomite rich upper section. Middle Jilh is comprised of a series of halite layers interbedded with anhydrite and dolomite (Jilh Salt) and overlying Jilh Dolomite. In the Burgan Field, a distinctive suite of six halite layers characterizes the lower part of the member and forms a distinctive seismic marker i.e. top Jilh Salt. These salts probably change into homotaxial anhydrite and dolomite facies in Umm Gudair and Sabriyah areas. The Jilh Dolomite (Sefidar Dolomite in Iran) is of demonstrable correlative value over a large area in the neighboring countries. Locally, a thin oolite development characterizes the lower part of the basal dolomite. The Upper Jilh consists of interbedded sequence of anhydrites and argillaceous dolostones over the Kuwait Arch and changes into a dominantly dolostone rich section in Dibdibba Trough.

The rock types derived from the petrophysical analysis clearly show the dominance of evaporites (Jilh E) on the regional highs, Burgan and NWRA (Fig. 5). This indicates a relatively shallower environment favoring the deposition of evaporites and based on this it can be inferred the potential areas for carbonate deposition will be away from these prominent highs at an optimal bathymetry. These can be on the structural highs with relatively lower relief and offstructure areas can also be an area of exploratory interest.

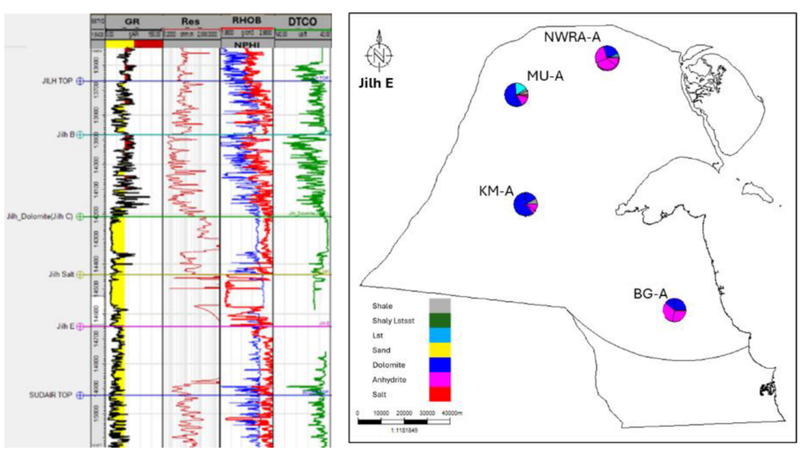

Figure 5—Log representation of Jilh section and the maps showing rock types derived based on petrophysical evaluation.

The Minjur Formation of Early Jurassic, Hettangian - Late Triassic, Rhaetian - Norian age represents the end of Triassic deposition and is marked by a brief return to clastic deposition as relative sea level fell at the end of the Triassic due to regional uplift of the Arabian Plate. The formation is composed of mixed siliciclastic and carbonate lithology. The formation is thickest and more sandstone -rich over the Burgan area and gradually thins and becomes shale dominated towards the north. The formation is divided into three members, Lower, Middle and Upper. Lower Minjur is composed of dark grey shales in the basal part, within it locally carbonate stringers are also developed. Within lower Minjur, the upper part is composed of dominantly sandstone with shale interbeds. The sandstone is fine to coarse grained and at times have red stained quartz grains. The shale is carbonaceous and locally transits to muddy coal as observed in South Kuwait wells. This unit becomes very thin in north Kuwait wells. The Middle Minjur is made up of two units; Minjur Clastics and overlying Minjur Lower Carbonates. Minjur Lower Carbonate consists of limestone (mudstone to wackestone) with associated bedded anhydrite and fine crystalline dolomite. Sandstone in the Minjur Clastic unit is the major reservoir in South Kuwait. The Upper Minjur is composed of multicolour, red, green, and grey blocky mudstone/shale overlain by interbedded dark grey calcareous and brown shale with limestone interbeds in North Kuwait wells. However, in South and West Kuwait, this unit is composed of mixed lithologies such as grey and brown shale, carbonaceous shale, sandstone, and limestone interbeds. Paleo-environments show a shallowing trend, with inner neritic conditions in the south and transitional conditions prevailing towards the north. In the south, distal conditions are interpreted for the uppermost

The rock types derived from petrophysical assessment suggest dominance of clay and shale as observed in the drilled wells (Fig. 6). Towards the western area in KM and MU a marked presence of carbonates is seen. Based on this, the clastic and the carbonate mixed environment exists more towards the west and can be attributed to the paleotopography which is favorable for carbonates as well. Towards the east on the regional highs, dominance of clastic can be inferred as a possibly proximity of the provenance for deposition of sandstone.

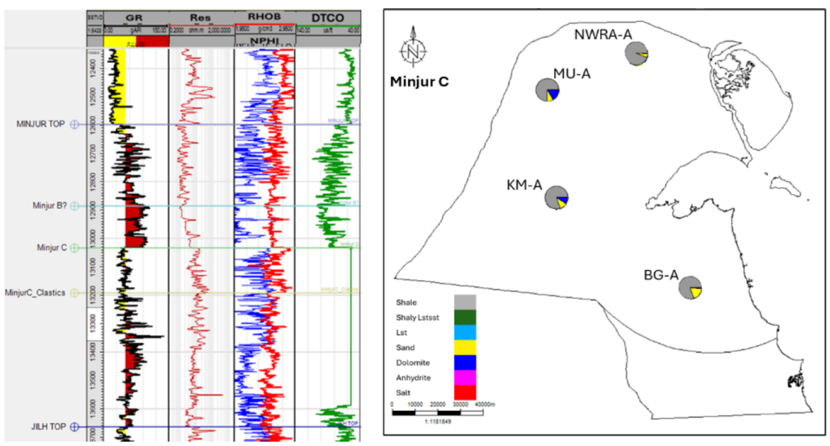

Figure 6—Log representation of Minjur section and the maps showing rock types derived based on petrophysical evaluation.

Seismic based Studies and Regional Analogues

Based on the regional understanding from well data, two arbitrary seismic sections (Fig.7 & 8) from the existing 3D volumes over multiple surveys were tied with the wells for further assessment. It was a challenge even to interpret the major formation tops from one well to another on this seismic. The imaging quality at the current depths makes it difficult to infer the thinning or wedging out of sequences. The Pre-Khuff formations of Permian age (not shown in the seismic profiles above) were also difficult to interpret. It was an easy conclusion that the Triassic-Paleozoic sequences in the seismic are affected with poor resolution, multiples and noise. The less penetration of energy further caused reduced signal to noise ratio affecting the proper imaging of primary events corresponding to Formation boundaries. The regional analogues indicate the occurrence of a complex geology in Triassic-Paleozoic sequences, and they might be contributing to the challenges in their adequate imaging by seismic.

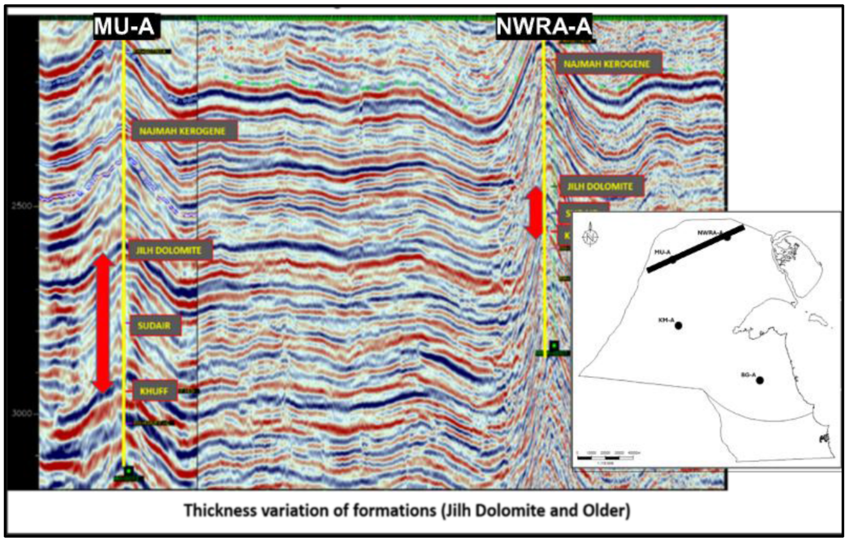

Figure 7—Arbitrary NW-SE seismic profile connecting wells MU-A and NWRA-A (inset map) showing Triassic to Jurassic sections.

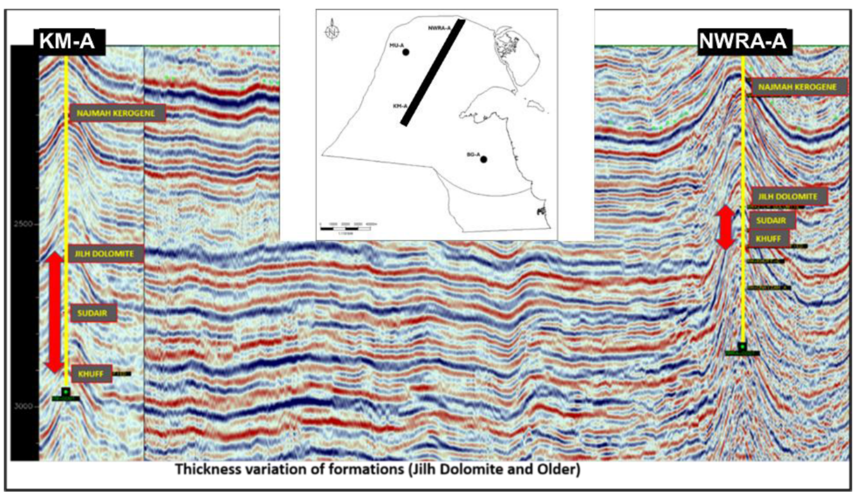

Figure 8—Arbitrary NW-SE seismic profile connecting wells KM-AAA and NWRA-AAA (inset map) showing Triassic to Jurassic sections.

An area where the thickness of the Triassic formations was higher and with available high-density full azimuth 3D PSTM seismic covering the deeper sections was chosen. A representative W-E profile (Fig. 9) shows the reflectors from Jurassic to Paleozoic. No stratigraphic features or well-defined structures in the deeper part could be seen in the seismic. The almost conformable nature of deeper reflectors is clear indication that these were infected with multiples.

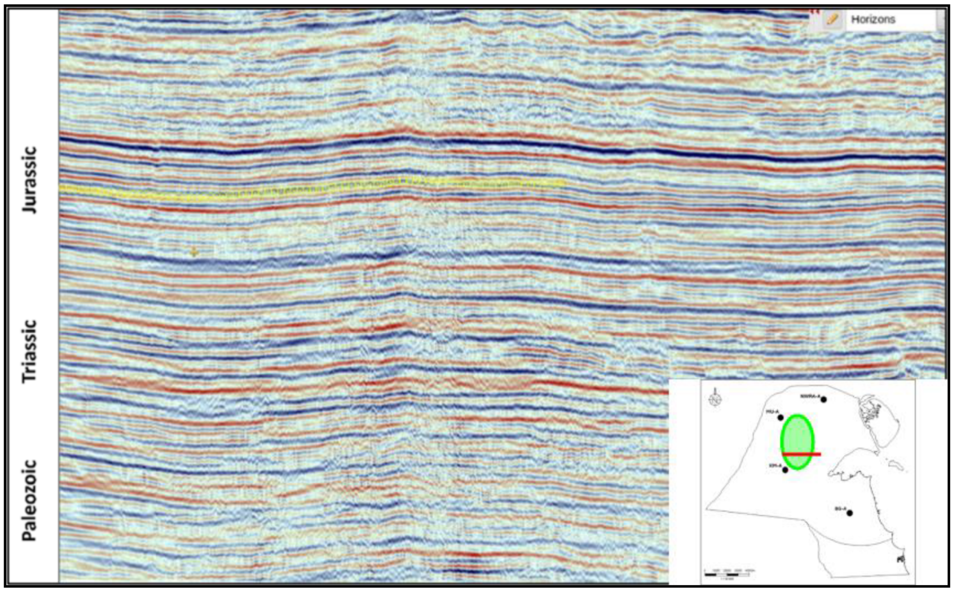

Figure 9—A representative W-E seismic profile from the 3D volume in the area having higher thickness of Triassic formations.

Many published technical papers from the region around Kuwait were investigated to see the nature of deeper events. Faqira et. Al 2009 brought out a very fruitful understanding of the Paleozoic sequences in The Kingdom of Saudi Arabia and in the Northwestern border of The State of Kuwait. The NW-SE seismic profile F-F′ through three drilled wells (Fig.10) very clearly shows the stratigraphic features and wedging out of the deeper sequences. This reference profile was very close to the Northwestern boundary of Kuwait, and it was a logical to assume that similar features and a similar geometry of the formations can be expected in Kuwait.

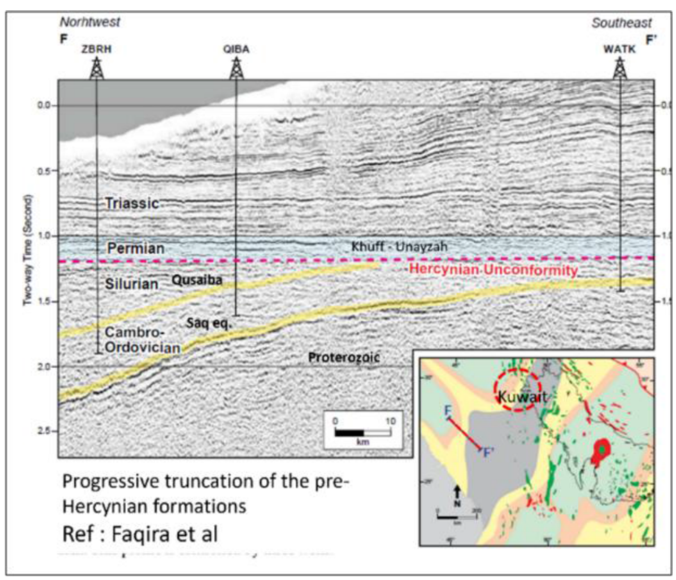

Figure 10—A NW-SE seismic profile from the Kingdom of Saudi Arabia.

Since the available PSTM (Fig. 11) over the area under study did not show anything like what was observed in the referred section, it would have been futile to decipher the sub-surface properties and different petroleum system elements using the available seismic. It was decided to re-process the 3D seismic in the depth domain. The full azimuth 3D depth domain reprocessed seismic data, that was processed separately for deeper and upper sections has brought out enhanced image of deeper sections and has produced a better image in the Triassic-Paleozoic in this area. New PSDM data (Fig.12) shows the clarity in deeper part and the enhancement in the Triassic sections with the appearance of dipping events in Paleozoic section that are wedging out.

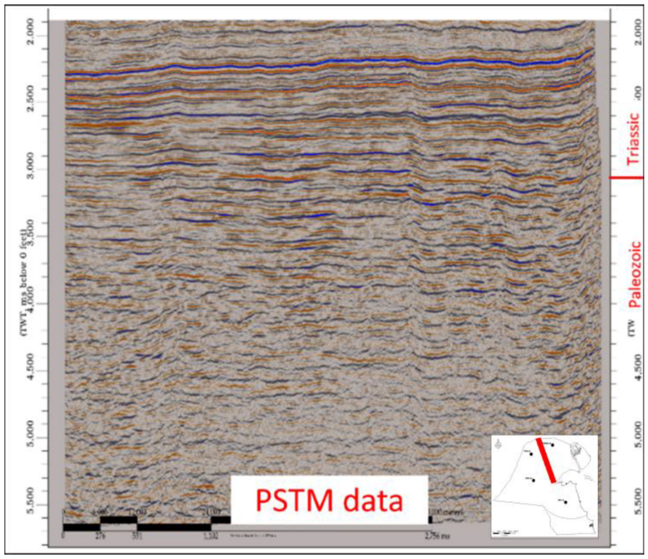

Figure 11—A NW-SE seismic profile from the PSTM seismic volume in the area under study.

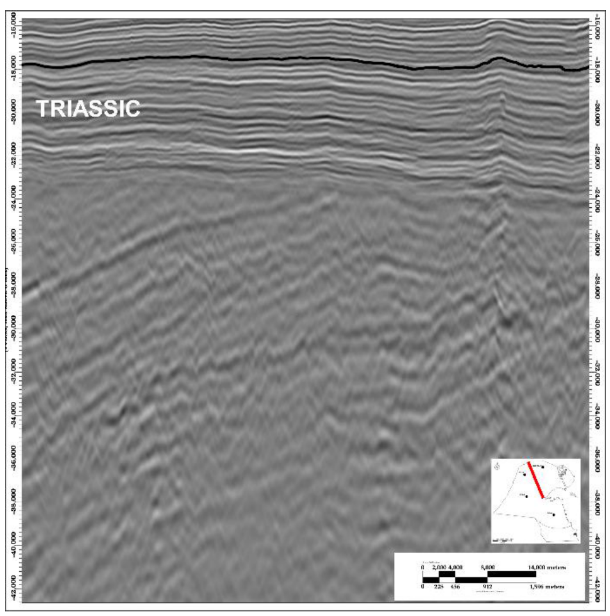

Figure 12—Northwest-Southeast seismic profile from reprocessed PSDM volume.

The NW-SE profile from the PSDM volume was compared with the reference seismic profile from The Kingdom of Saudi Arabia. A striking similarity was observed in the Paleozoic sequences between these two profiles (Fig. 13). The Pre-Khuff Hercynian unconformity was very well brought out and was matching with the interpretation in the reference profile. This led to enhanced confidence in the sub-surface image after depth processing.

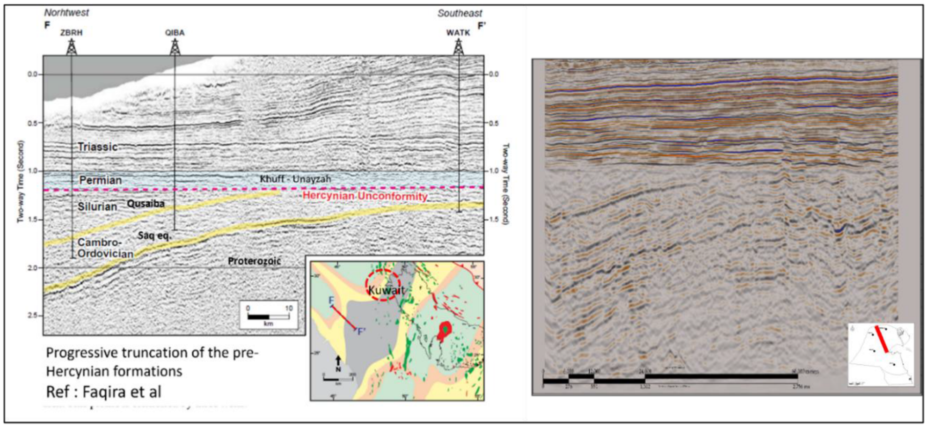

Figure 13—NW-SE seismic profile F-F′ lying in The Kingdom of Saudi Arabia and from reprocessed PSDM volume in Kuwait.

The deepest well in Burgan area, that was drilled through almost the entire Paleozoic sequences reaching technical/economic basement, could not conclusively establish the presence of Silurian Qusaiba formation which is expected to be the source of hydrocarbons for deeper reservoirs. The seismic profile from PSDM volume indicates wedging out of some deeper sequences. The preliminary conclusion was that Silurian section was either not deposited or eroded in over the paleo-structural highs. In regional context Silurian Akash formation, time equivalent also has good source potential especially in the northern part of Kuwait, however its' areal extent is less understood. The new PSDM threw new light on ultra deep formations in addition to providing an enhanced imaging of the Triassic sequences.

Conclusion

The Minjur, Jilh and Sudair formations, representing Triassic section in Kuwait, are envisaged to have significant potential for nonassociated gas and condensates. The exploration of these formations is constrained by insufficient well-based subsurface information and seismic imaging quality. Abdullah Alkandari et. al (2023) have described four plays, bounded by prominent top and base seals, within Triassic section, Sudair, Lower Jilh, Middle Minjur - Upper Jilh and Upper Minjur plays. Exploratory efforts have yielded successful results in Lower Jilh and it is considered as the most promising play. In regional context Minjur is successfully explored in the region.

Assessment of the basement architecture and the tectono-stratigraphy, along with the development of the paleo depositional environment and analysis of the regional context of the Triassic-Paleozoic on the Arabian Plate may identify potential reservoir strata and entrapments in Kuwait. The N-S trending Kuwait Arch and the reactivation of structural grain inherited from Hercynian and older tectonism has significant impact on the presence of the deeper strata. The existence of intra-basinal highs with the possibility of thick Paleozoic source rock deposition in the adjacent basinal lows can form the potential locales for exploration. Preliminary structural analysis indicates these are older structures controlled by basement tectonics. Furthermore, it is understood that late Jurassic time is the peak migration period for deeper hydrocarbon sources and any structuration that predates this period can provide the necessary entrapment in these structures.

Modern technologies, the latest processing software, the availability of more regional information based on published technical papers on related matter and innovative exploration practices were implemented to resolve the challenges associated with the Triassic- Paleozoic in The State of Kuwait leading to a more refined subsurface model. The wells drilled on the identified prospects will provide direct information of the ultra-deep reservoirs, entrapments, source etc. and will be used for fine tuning the present studies and for future prospect identification.

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